Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

18

Data Columns

37

Missing Ratings

Imputed via category median

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Size

Behavior

Discounts, Frequency, Ratings, Shipping

Data Preparation in Python

→ Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

→ Clean & Standardize

Imputed missing ratings; renamed columns to snake_case

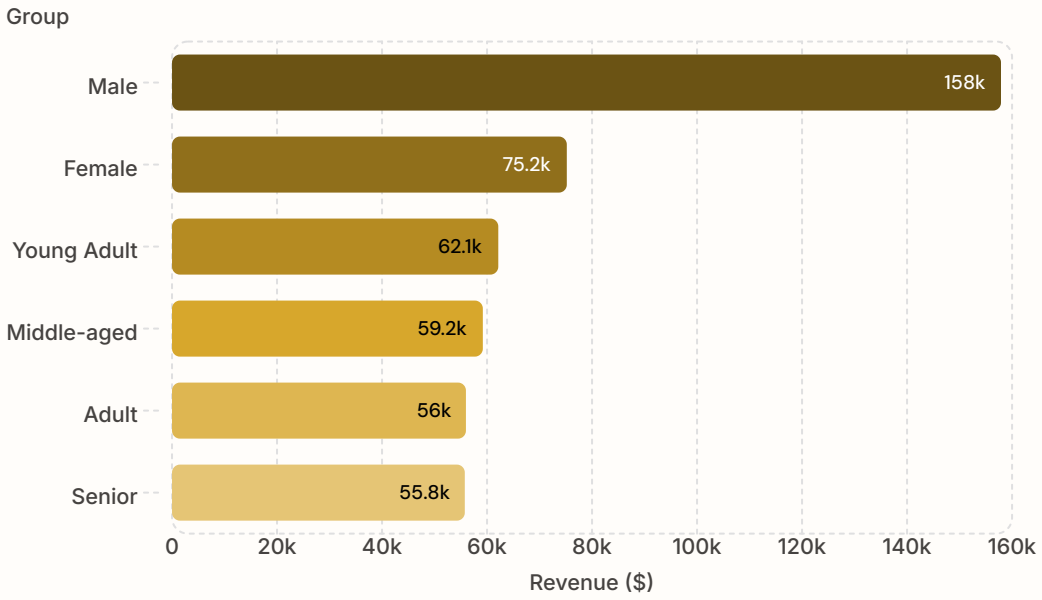
→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ Database Export

Loaded cleaned data into MySQL Workbench

Revenue Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults lead across all age groups
- Express shipping users avg. **$60.48** vs. Standard **$58.46**

Top-Rated Products

1

Gloves

Avg. rating **3.86**

2

Sandals

Avg. rating **3.84**

3

Boots

Avg. rating **3.82**

4

Hat

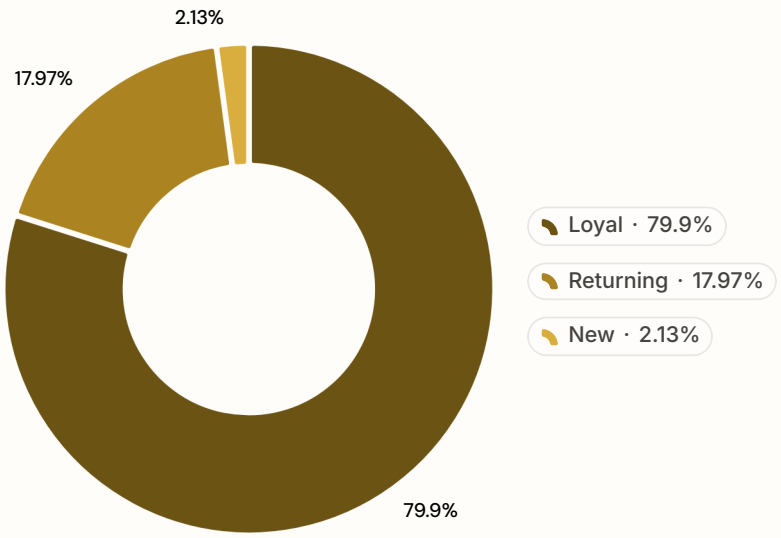
Avg. rating **3.80**

5

Handbag

Avg. rating **3.78**

Customer Segmentation



Segments by Purchase History

Loyal — 3,116

Majority of the customer base

Returning — 701

Engaged but not yet loyal

New — 83

Acquisition opportunity

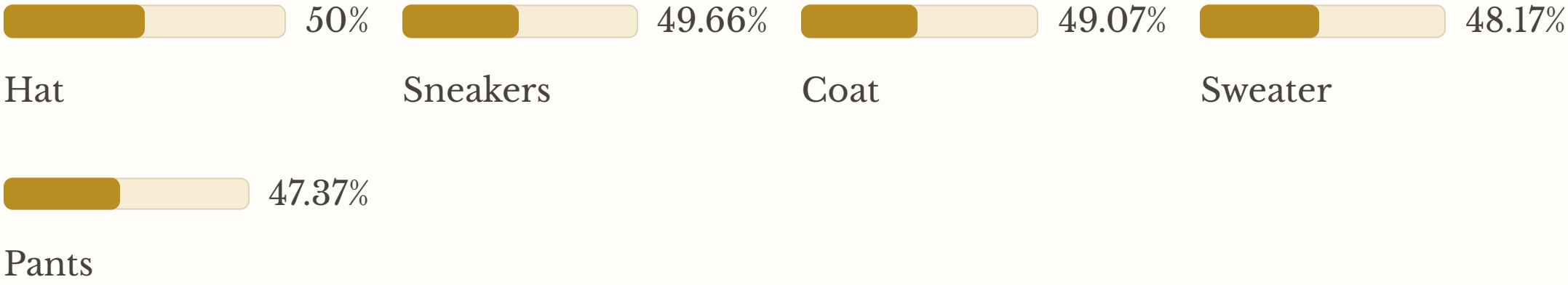
Subscriptions & Discounts

Subscribers vs. Non-Subscribers

Status	Customers	Avg. Spend	Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Repeat buyers (>5 purchases): **958** subscribed vs. **2,518** not subscribed.

Most Discount-Dependent Products



Top Products by Category

Accessories

1. Jewelry — 171 orders
2. Sunglasses — 161
3. Belt — 161

Clothing

1. Blouse — 171 orders
2. Pants — 171
3. Shirt — 169

Footwear

1. Sandals — 160 orders
2. Shoes — 150
3. Sneakers — 145

Power BI Dashboard

Key Metrics

Filters: Subscription, Gender, Category, Shipping Type. Visuals: revenue & sales by category and age group.

3.9K

Customers

$59.76

Avg. Purchase

3.75

Avg. Rating

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers; move them to "Loyal"

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users